

Technical Document

Key Challenges & Solutions

1. Real-Time Data Fetching from Yahoo Finance

- Challenge: Getting live stock market data without exposing API keys or using paid services.

- Solution: Used the ``yahoo-finance2`` NPM package, which allows fetching real-time stock and crypto data without requiring API keys. This enabled secure and efficient integration of financial data directly in the frontend/backend.

2. Rate Limiting & Data Accuracy

- Challenge: Avoiding request throttling or outdated data due to too many or too few fetches.

- Solution: Implemented strategic data polling intervals and error handling to gracefully manage data refresh without overwhelming the API or introducing lags.

3. User-Friendly UI for Financial Information

- Challenge: Displaying complex financial data in a clear, intuitive way for both novice and experienced users.

- Solution: Built a clean UI using React and TailwindCSS, highlighting key indicators like price, change %, volume, etc., with clear color-coding and responsive layout.

4. Environment Variable Elimination

- Challenge: Common financial APIs require API keys, usually stored in environment variables, complicating deployment.

- Solution: Chose a package (``yahoo-finance2``) that eliminates the need for API keys or environment variables, simplifying development and deployment workflows.

5. Error Handling and Robust UX

- Challenge: Handling API downtimes, ticker mismatches, or data parsing errors gracefully.

- Solution: Added fallback messaging, loader states, and alerts so users are never left without feedback when something fails.